

Department of Cyber Security

2025-2028- Odd Semester

III Year /V Sem

CS23533-FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

Zeroth Review :Zeroth Review

PROJECT TITLE: Intelligent Digital Evidence Reconstruction System

Batch Members :

1.Name: Sachin Krishna B

Reg.no: 241901507

2.Name: Arilli Jagadeesh A

Reg.no: 241901502

Supervisor:

Dr. S.Bhuvana

Assistant Professor[SG]

OUTLINE

- Abstract
- Objective
- Literature Survey
- Summary of Literature
- Proposed System & Novelty
- Block Diagram
- Hardware/Software requirements
- References

ABSTRACT

- Cyber crimes are increasing rapidly, making digital investigations more challenging.
- Investigators collect evidence from multiple sources such as system logs, authentication logs, USB logs, browser history, and network traffic.
- Manual investigation is time-consuming and may miss important evidence.
- CyberTrace AI is an AI-powered digital forensic system that automatically collects, parses, correlates, and analyzes digital evidence.
- The system reconstructs the sequence of cyber events and generates an investigation timeline.
- It assists investigators in identifying suspicious activities quickly and accurately.
- The solution reduces investigation time and improves evidence analysis.

OBJECTIVE

- Develop an intelligent digital forensic investigation system.
- Automatically collect digital evidence from different log sources.
- Parse and normalize heterogeneous log formats.
- Correlate events from multiple evidence sources.
- Reconstruct the attack timeline.
- Detect suspicious user activities using AI.
- Visualize the investigation process.
- Generate automated forensic reports.

Literature Survey

TITLE	YEAR	AUTHOR	TECHNIQUE
Intelligent Digital Forensics Using Machine Learning	2022	M. Alenezi et al.	Machine Learning-based Digital Evidence Analysis
Machine Learning-based Digital Evidence Analysis	2021	S. T. Zargar et al.	Event Correlation & Log Analysis
Automated Timeline Reconstruction in Digital Forensics	2023	J. Kim et al.	Timeline Reconstruction using AI
AI-Based Cyber Attack Detection and Investigation	2024	A. Sharma et al.	Artificial Intelligence & Anomaly Detection

EXISTING SYSTEM

- Manual log analysis
- Time-consuming investigation
- Separate analysis of each evidence source
- Difficult to identify attack sequence
- Limited visualization
- No intelligent evidence reconstruction
- High dependency on forensic experts

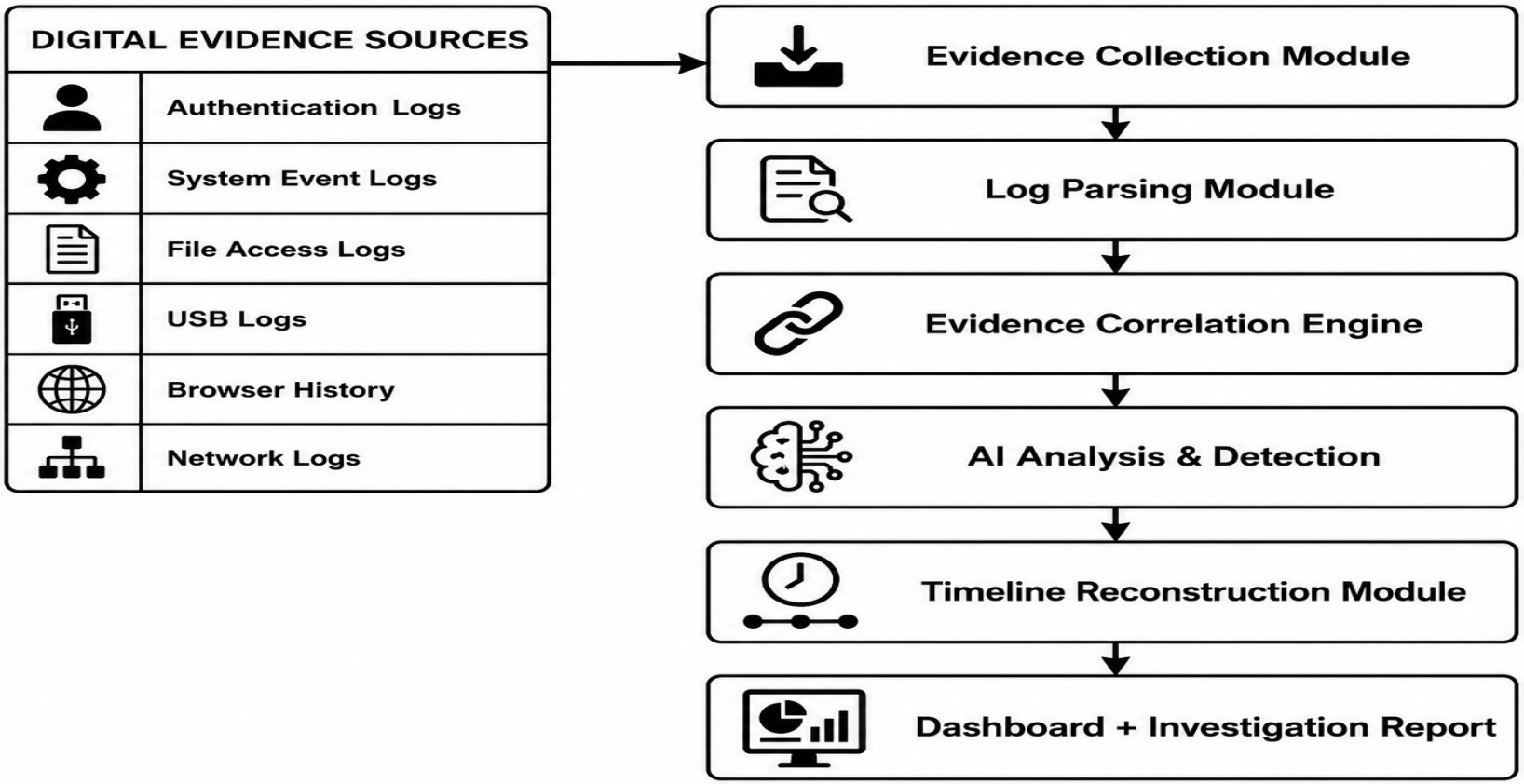
PROPOSED SYSTEM

- CyberTrace AI automatically collects digital evidence from multiple sources, correlates events using AI techniques, reconstructs the cyber incident timeline, and generates visual forensic reports.

Novelty in Proposed System

- AI-based evidence analysis
- Automatic log parsing
- Multi-source evidence correlation
- Intelligent attack timeline reconstruction
- Interactive investigation dashboard
- Risk score calculation
- Automated report generation
- Graph visualization of attack sequence

BLOCK DIAGRAM(PROPOSED SYSTEM)



Hardware Requirements

Hardware Requirements	
Component	Specification
Processor	Intel Core i5 / AMD Ryzen 5
RAM	8 GB Minimum
Storage	256 GB SSD
Internet	Required
Monitor	1366×768 or higher

Software Requirement

Software Requirement	
Frontend	HTML, CSS, JavaScript, React.js
Backend	Python, Flask,
Database	SQLite / PostgreSQL
AI & ML	Scikit-learn,Pandas, NumPy
Visualization	Chart.js, Plotly
Development Tools	VS Code, Git, GitHub
Operating System	Windows 10/11,Ubuntu Linux

References

- Casey E., Digital Evidence and Computer Crime.
- NIST SP 800-86 – Guide to Integrating Forensic Techniques.
- OWASP Security Documentation.
- IEEE Digital Forensics Research Papers.
- ACM Digital Library.
- Python Documentation.
- Flask Documentation.
- Scikit-learn Documentation.
- MITRE ATT&CK Framework.
- Cybersecurity and Infrastructure Security Agency (CISA).